

INSuRE Spring 2023 Project Proposal

Data Provenance in a Topic Focused Small Language Model

Sponsor: National Security Agency

Mark Boutwell

National Security Agency – NSA Research Scientist -
maboutw@radium.ncsc.mil

Brian Ricks

UTDALLAS- Faculty - bwr031000@utdallas.edu

Dhyana Sai Reddy Dalli – UTDALLAS – dx210031@utdallas.edu

Subhash Nirmal Kolluru – UTDALLAS – sxk210036@utdallas.edu

Aryavardhun Guntupally – UTDALLAS – axg210117@utdallas.edu

Yaswanth Kulas Devarasetty – UTDALLAS – yxd210007@utdallas.edu

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1) PROJECT SUMMARY:

The project aims to develop an in-depth understanding of data provenance in a Topic Focused Language Model (TFLM) using small, targeted datasets. The main objectives of the project can be summarized as follows:

1. **Understanding Data Provenance:** The project will delve into the origins and lifecycle of data within a TFLM. This involves tracking the data from its source through its processing and eventual use in the model.
2. **Use of Small Targeted Datasets:** The project will utilize small, specific datasets to train the TFLM. These datasets are chosen for their relevance to the topic of focus.
3. **Pilot Demonstrations:** The project will conduct pilot studies to demonstrate one or more frameworks. These pilots serve as practical applications of theoretical concepts and provide tangible evidence of the model's capabilities.

In essence, this project is a comprehensive exploration of how data is sourced, processed, and utilized in a TFLM, with an emphasis on practical demonstrations through pilot studies.

Keywords: Data Provenance, Topic Focused Language Model (TFLM), Small Targeted Datasets, and Pilot Demonstrations.

Terminology:

1. **Data Provenance:** The historical record and origin of data, including its sources, transformations, and handling throughout its lifecycle.
2. **Topic Focused Small Language Model (TFLM):** A compact and specialized language model designed to excel in understanding and generating text on specific subjects or topics.

Responsibilities:

We as a team would learn the basics of how to analyze the TFLM model. Then the remaining of us will work to identify what kind of parameters generally are targeted and resolve and optimize the model.

Estimated Project Budget:

This project requires approximately 500 hours of work and has no monetary requirements.

Deliverables:

Key Deliverables	Due Date
Project Proposal	9/8
Phase One	9/30
Midterm Presentation	10/6
Phase Two	10/30
Phase Three	11/10
Phase Four	11/20
Final Presentation and Report	12/08

Executive Summary

This project seeks to enhance data traceability within specialized Topic Focused Small Language Models (TFLMs). Through rigorous analysis, experimentation, refinement, and comprehensive documentation, we aim to contribute valuable insights and recommendations for improving data provenance in TFLMs, fostering responsible AI development across diverse domains.

2) MOTIVATION:

While Federated Learning is designed to preserve data privacy, there are still some privacy concerns that need to be addressed. These include:

Model Inversion Attacks: In this type of attack, an adversary can reverse engineer the model parameters to gain access to the training data. This type of attack involves using the federated model's predictions to infer sensitive information about the training data, potentially compromising the privacy of the participants.

Model Stealing Attacks: Adversaries can steal the trained model and use it for malicious purposes. In this type of attack, an adversary can extract the model's parameters after it has been trained on multiple parties' data, allowing them to access private information from the data.

Privacy Leaks through Model Parameters: The model parameters, which are shared during Federated Learning, can reveal information about the training data.

Unwanted data collection: Federated Learning requires data from multiple devices to be transmitted to a central server, which can lead to unwanted data collection and storage.

Model Biases: The model may learn biased representations due to the distribution of data on the devices.

Data Poisoning: An adversary can manipulate the training data on a device to poison the model parameters, which can lead to incorrect or malicious results.

To address these privacy issues, Federated Learning needs to be combined with privacy-preserving techniques such as differential privacy and secure multi-party computation to ensure that the training data is kept confidential and that the model parameters are kept secure.

Are there any more possible attacks on Federated Learning systems.

Yes, there are several other possible attacks on Federated Learning systems, some of which include:

Model Poisoning: This type of attack involves introducing malicious updates to the federated model by an adversary, causing the model to behave abnormally and make incorrect predictions.

Model Replacement: An adversary can replace the federated model with their own, potentially compromising the privacy of the participants or manipulating the model's predictions.

Backdoor Attacks: This type of attack involves inserting malicious code into the model, which allows an attacker to compromise the privacy of the participants or manipulate the model's predictions.

These attacks highlight the importance of robust privacy and security measures in Federated Learning systems, such as differential privacy and secure multi-party computation.

3) LITERATURE REVIEW:

Li, X., & Fourches, D. (2021). SMILES pair encoding: a data-driven substructure tokenization algorithm for deep learning. *Journal of chemical information and modeling*, 61(4), 1560-1569.

Introducing SMILES Pair Encoding (SPE), a novel tokenization method for improving deep learning models in cheminformatics. SPE learns common SMILES substrings from extensive chemical data, using this knowledge to tokenize SMILES for model training. Unlike standard atom-level tokenization, SPE incorporates human-readable and chemically meaningful SMILES substrings as tokens. Our studies demonstrate SPE's superior performance in molecular generation and quantitative structure-activity relationship (QSAR) prediction tasks, outperforming atom-level tokenization in generating diverse structures while maintaining data fidelity. Across 24 benchmark datasets, SPE consistently matches or outperforms other tokenization methods, making it a promising choice for SMILES-based deep learning models.

Li, Z., Jiang, M., Wang, S., & Zhang, S. (2022). Deep learning methods for molecular representation and property prediction. *Drug Discovery Today*, 103373.

In recent years, AI has greatly advanced computer-aided drug design (CADD). This review covers how deep learning methods are applied to represent molecules and predict their properties, categorizing these methods based on the type of molecular data they handle. We also discuss key models like ensemble learning and transfer learning and consider the interpretability of their results. Finally, we touch upon the challenges and opportunities in using deep learning for these tasks in CADD.

Suprem, A., Vaidya, S., Venugopal, A., Ferreira, J. E., & Pu, C. (2022). EdnaML: A Declarative API and Framework for Reproducible Deep Learning. *arXiv preprint arXiv:2211.06783*.

Machine learning is driving advancements in text, images, videos, and audio processing. In real-world applications, various models with different settings are used, making hyperparameter tuning crucial. Additionally, the data collection and processing aspects of ML pipelines are gaining importance for creating robust classifiers. EdnaML is introduced as a framework with a user-friendly API for reproducible deep learning. It offers both low-level building blocks for manual configuration and a high-level API for automating data tasks, classifier training, deployment, and monitoring. EdnaML simplifies managing ML pipelines at higher abstraction levels while allowing flexibility for customization. Several examples, including a fake news classification system with six sub-pipelines, demonstrate EdnaML's capabilities.

4) SPECIFIC AIMS:

Explore and analyze data provenance techniques in Topic Focused Small Language Models (TFLMs), customize and refine these techniques for TFLMs, rigorously evaluate their effectiveness through pilot experiments, maintain comprehensive documentation, validate, and finalize the data provenance framework, and contribute valuable insights and recommendations for enhancing data provenance in TFLMs.

5) APPROACH:

Phase 1: Analyze studies with frameworks that have a direct or indirect focus in data provenance and demonstrate techniques such as machine learning pipelines, classifiers, and model monitoring. The goal is to explore data provenance techniques using small targeted datasets in a TFLM to gain awareness and understanding and identify studies with functional candidate data provenance techniques.

Phase 2: Identify five candidate studies from Phase I for evaluation. Conduct quick pilot(s) to reproduce the selected studies. In the first set of pilots(s), use the original study's framework and dataset. In the second set of pilots(s), customize the original frameworks and develop small targeted datasets. Compare the findings from the original and custom frameworks and datasets. Define success criteria in the context of using small targeted datasets and implications to data provenance.

Phase 3: Assess all of the pilot findings and create two refined frameworks and small targeted datasets. The goal is to provide a comprehensive demonstration of the refined frameworks and datasets. Provide recommendations on a final framework and dataset. Document the new frameworks, datasets, respective workflows, performance monitoring, and so on.

Phase 4: Test and evaluate the findings from Phase I-III to develop and prove a final framework and small targeted dataset. Outline the dataset provenance, model monitoring, and other relevant findings and outcomes.

Dataset Description:

The provided data is structured with columns, each serving a specific purpose in describing individual entries. The "ID" column likely contains unique identifiers for each entry, facilitating the distinct identification of records. Meanwhile, the "TITLE" column appears to hold concise descriptions or names associated with each entry, offering a quick reference point. The "ABSTRACT" column provides more in-depth information, supplying detailed summaries or descriptions beyond the title.

The subsequent columns, namely "Computer Science," "Physics," "Mathematics," "Statistics," "Quantitative Biology," and "Quantitative Finance," seem to represent categorical variables. These columns likely contain binary values (0 or 1), indicating the presence or absence of each entry's association with the corresponding academic field. For instance, the "Computer Science" column may denote whether an entry is related to Computer Science, while the "Physics" column indicates its connection to the field of Physics, and so on.

This structured format suggests that the dataset is designed to showcase the interdisciplinary nature of the entries, with each one potentially having relevance to multiple academic fields. The binary values in the disciplinary columns enable a straightforward identification of the academic domains with which each entry is associated. This organization facilitates efficient data analysis and classification based on the specified academic categories, offering insights into the diverse intersections between Computer Science, Physics, Mathematics, Statistics, Quantitative Biology, and Quantitative Finance within the dataset.

Data Cleaning and EDA:

The provided code incorporates a comprehensive set of text preprocessing techniques. Initially, the text is converted to lowercase, contractions are expanded, and apostrophes are removed. Subsequently, stemming and lemmatization are applied to reduce words to their root forms, promoting text normalization. The process also involves the removal of stop words, common words that do not contribute significantly to the meaning of the text. Finally, non-word characters and extra whitespaces are eliminated to ensure a clean and standardized text representation. These combined steps aim to enhance the quality of the text data for subsequent analysis or modeling.

During our exploratory data analysis, we generated multiple plots to gain insights into the dataset. From this set of plots, we carefully selected two that offer particularly meaningful information about the data. The first plot(fig 1(a)) provides a detailed view of the distribution of data points concerning average word count. In this representation, the X-axis denotes the varying word counts, while the Y-axis reflects the corresponding number of abstracts associated with each specific word count.

The second plot(fig 1(b)), on the other hand, delves into the dataset's composition by illustrating the quantity of records linked to each label. This visual representation on the graph effectively communicates the distribution and prevalence of different labels within the dataset. Together, these two plots contribute to a comprehensive understanding of the data, highlighting patterns related to word counts and providing insights into the distribution of records across various labels.

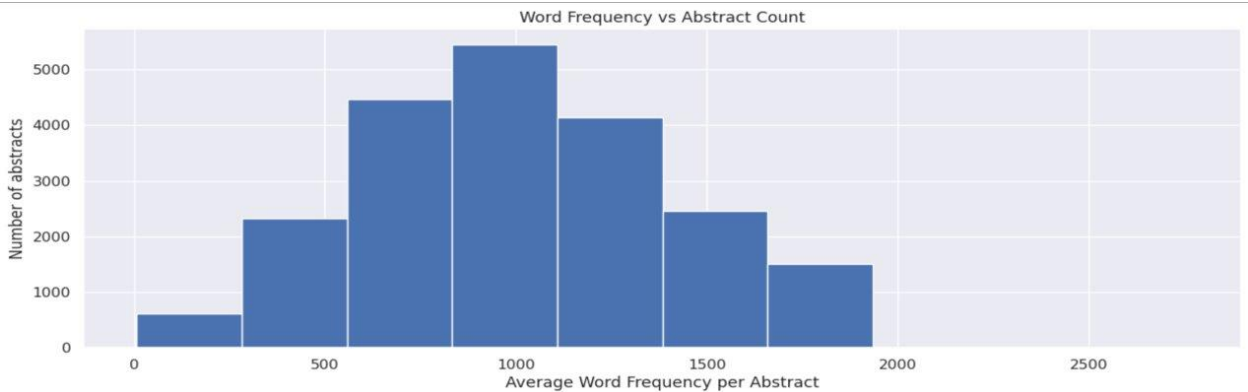


fig 1(a)

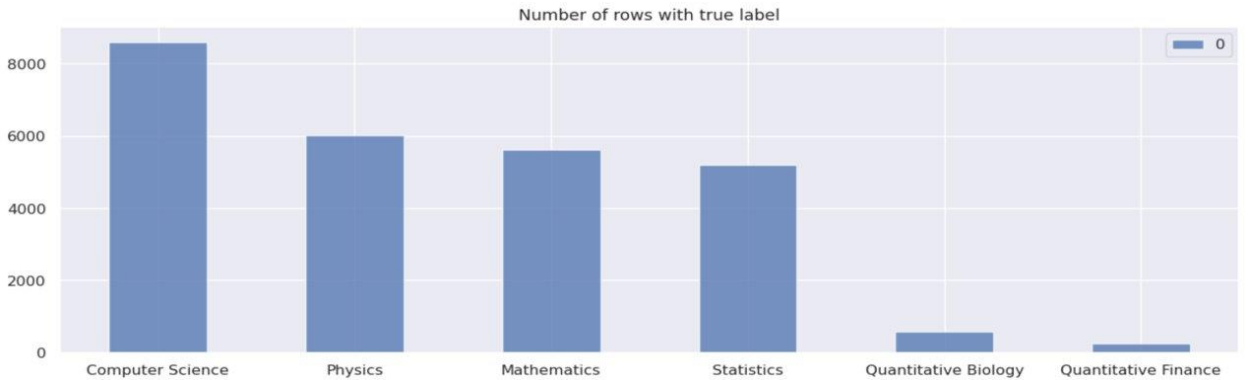


fig 1(b)

Model Implementation:

We have logistic regression models trained for each specified label in the variable. Each model is fitted using training data, and predictions are generated for both the training and testing datasets. The logistic regression coefficients are then examined to identify the top words associated with each label, selecting them based on their impact on the model. These top words are stored in a dictionary for future reference.

Moreover, we evaluate the performance of the logistic regression models by computing accuracy scores. These scores are calculated for both the training and testing sets, providing insights into the models' ability to generalize to new data. The results are systematically organized into a dataframe, consolidating accuracy scores for different labels across both training and testing datasets.

	train_accuracy	test_accuracy
Computer Science	90%	86%
Physics	95%	93%
Mathematics	92%	90%
Statistics	91%	88%
Quantitative Biology	98%	97%
Quantitative Finance	99%	99%

Following the model generation, we extract feature importance for each target variable. Subsequently, we identify the top six significant keywords that exert influence on the model, as indicated in the accompanying table.

	1	2	3	4	5	6
Computer Science	Language	Robot	Complexity	Paper	Software	Robots
Physics	Spin	Physics	Optical	Electron	Temperature	Phase
Mathematics	Prove	Mathbb	Asymptotic	Group	Theorem	Spaces
Statistics	Learning	Statistical	Sample	Bayesian	Regression	Likelihood
Quantitative Biology	Brain	Cells	Cell	Population	DNA	Biological
Quantitative Finance	Market	Portfolio	Volatility	Price	Risk	Stock

6) ANTICIPATED ISSUES AND CHALLENGES:

As most of us are beginners in the field of Machine Learning and its security applications, tackling big data sets and understanding the models would be a hurdle initially.

The project may face challenges including data availability, the complexity of customizing techniques for TFLMs, resource intensity, defining appropriate evaluation metrics, maintaining comprehensive documentation, ethical considerations, integration complexities, staying current in a rapidly evolving field, effective team coordination, and unforeseen external factors, all of which will require careful management and strategic solutions for successful project execution.

7) BROADER IMPACT:

This project's broader impact extends to the realms of both technology and society. By enhancing data provenance in Topic Focused Small Language Models (TFLMs), it lays the foundation for more transparent, accountable, and reliable AI systems. This not only bolsters the trustworthiness of AI applications but also has far-reaching implications across industries, including healthcare, finance, and research, where data traceability and ethical considerations are paramount. Moreover, the documentation and best practices emerging from this project have the potential to inform future AI research, nurturing a culture of responsible AI development and deployment, ultimately benefiting society at large.

8) INVESTIGATOR BIOGRAPHIES:

Dhyana Sai Reddy Dalli holds a bachelor's degree in Electronics and Communications Engineering and is doing his masters at The University of Texas at Dallas, graduating in December 2023 with a focus in Data Science. He has experience working at TCS as a System Engineer. He has done Machine Learning as an undergraduate course and is looking forward to gaining expertise in the field of Data Science. He also has experience in Python, Java and R.

Subash Nirmal Kolluru, a seasoned data scientist with expertise in machine learning. My experience at Tiger Analytics, where I served as a Senior Data Analyst, includes collaborating on projects that increased company profits through accurate predictions, optimizing customer acquisition strategies, and implementing automation solutions..

Aaryavardhun Guntupally is pursuing his masters at The University of Texas at Dallas, graduating in Dec 2023 with a focus in Intelligent Systems. He started his career in Innomed 3D Solutions as a Software Engineer with 2+ years of experience. He Posses Web Development Skills as well as Programming skills such as Python, Java, SQL and Matlab.

Yaswanth Kulas Devarasetty is doing his masters at The University of Texas at Dallas with a focus in Intelligent Systems. He started his career in Wipro Technologies as a Project Engineer with 3+ years of experience. He Posses full stack experience with Technologies like React, typescript, MySQL, ASP .Net and graphQL.

9) Progress:

In our research project, we utilized a dataset sourced from Kaggle, which was primarily structured for classification purposes. Our initial step involved thorough data cleaning procedures to enhance the quality and integrity of the dataset. This process entailed removing inconsistencies, missing values, and other anomalies to ensure the dataset's reliability. Subsequently, we conducted an in-depth analysis of the dataset, studying its underlying patterns, characteristics, and potential insights. Our aim was to gain a comprehensive understanding of the data, which served as a foundational step to prove Data Provenance.

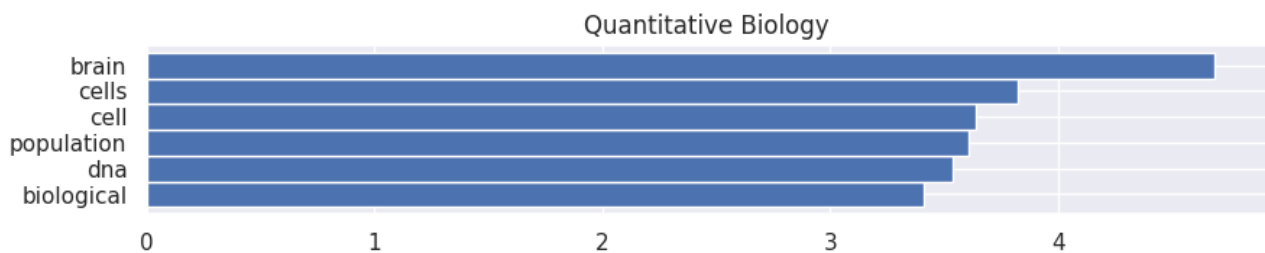
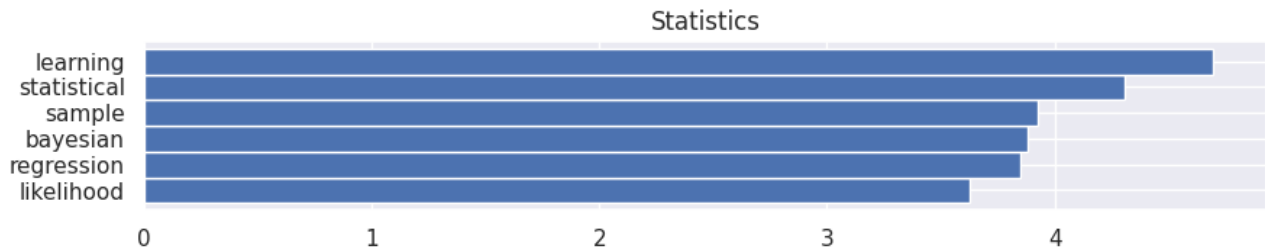
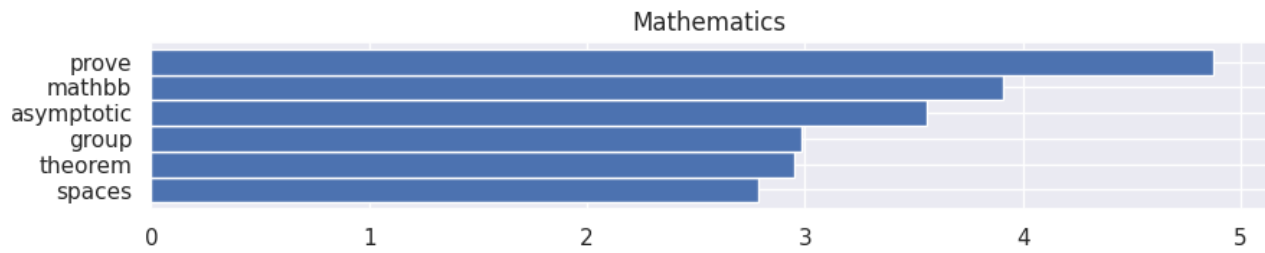
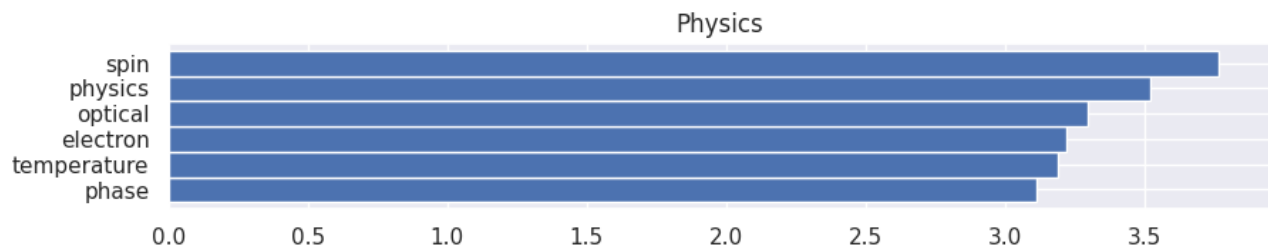
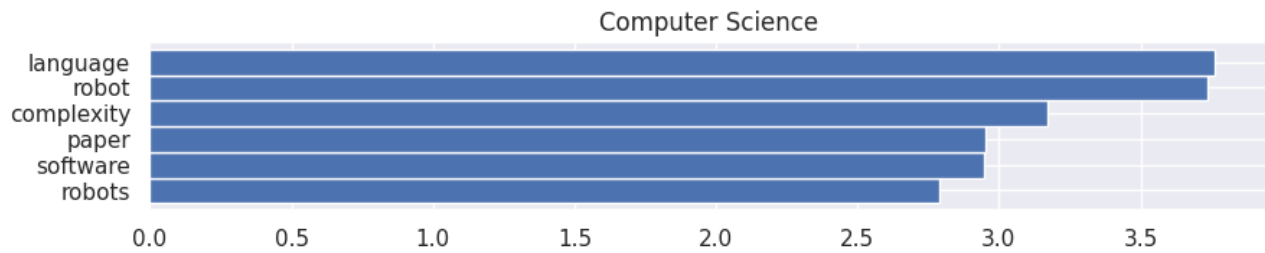
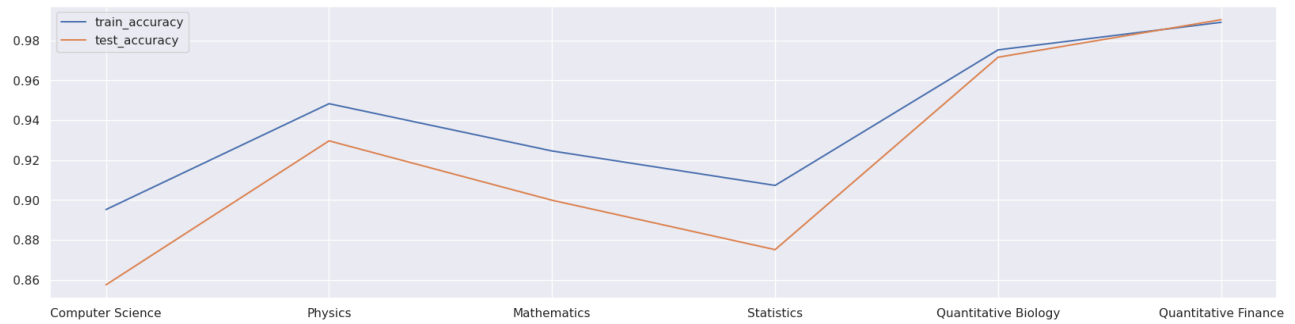
Key Challenges

Data Preprocessing: Text data often requires extensive preprocessing, including tasks such as tokenization, stop word removal, stemming or lemmatization, and dealing with special characters. This preprocessing can be time-consuming and may vary based on the language and the specific dataset.

Natural Language Understanding: Understanding the semantics, context, and sentiment of text is complex. Building accurate models for natural language processing (NLP) tasks, like sentiment analysis or named entity recognition, can be challenging.

Data Imbalance: Text classification problems often suffer from class imbalance issues, where certain categories or labels may have significantly more or less data than others, making it harder to train balanced models.

10)PLOTS:



11) FUTURE WORK:

1. Advanced ML Models (Deep Learning):

- Explore the integration of advanced machine learning models, specifically deep learning techniques.
- Deep learning models, with their complex structures, may enhance the accuracy and capabilities of small language models.

2. Hyperparameter Experimentation:

- Conduct experiments with different hyperparameters.
- Adjusting hyperparameters, such as learning rates or layer configurations, can significantly impact model accuracy.

3. Diverse Dataset Testing:

- Test the models on a variety of datasets.
- Evaluating performance across diverse datasets ensures the model's adaptability and generalization.

4. Encryption Techniques for Security:

- Implement and experiment with various encryption techniques.
- Explore encryption methods to ensure secure and safe transactions, especially important for sensitive data handling.

12) REFERENCES:

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